

P. Shubham Parashar

Theoretical condensed matter physics | quantum transport | kinetic theory | electron hydrodynamics

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RESEARCH PROFILE

I develop analytical and computational descriptions of transport in spatially inhomogeneous two-dimensional electron systems, with an emphasis on what finite-resolution measurements can identify. Current work connects semiclassical Boltzmann kinetics, electron hydrodynamics, magnetotransport, quantum Hall response, symmetry-selected kinetic channels, and controlled higher-gradient/nonlocal corrections.

SELECTED TECHNICAL CONTRIBUTIONS

- **Kinetic-to-hydrodynamic transport:** spatial magnetotransport around controlled inhomogeneities; systematic tests of stress-level closure; symmetry-selected local nulls that isolate higher kinetic harmonics.
- **Finite-resolution inference:** explicit visibility/identifiability tests for thermoelectric Hall contrast and viscous response in scanning-probe measurements.
- **Experiment-facing theory:** transport modeling tied to nanoscale potentiometric imaging of graphene p-n junctions and quantum Hall observables.

EDUCATION AND RESEARCH APPOINTMENTS

2022-2026	Ph.D., Physics, University of California San Diego - Advisor: Michael M. Fogler
2017-2022	Researcher, University of California San Diego - research with Benjamin Grinstein and Congjun Wu
2016-2017	M.S., Physics, Indian Institute of Science - Advisor: Rahul Pandit
2012-2016	B.S., Physics, Indian Institute of Science

PUBLICATIONS AND CURRENT MANUSCRIPTS

1. **Complete kinematic null for local kinetic dissipation in a sixfold-driven electron fluid.** P. Shubham Parashar. *Under review at Physical Review B* (2026).
2. **Fourth-order closure obstruction and chiral nonlocality in circular kinetic magnetotransport.** P. Shubham Parashar. *Under review at Physical Review B* (2026). [arXiv:2607.22730](#).
3. **Thermal and viscous contrast in quantum Hall scanning-probe images.** P. Shubham Parashar. *Submitted to Physica E* (2026). [arXiv:2607.16315](#).
4. **Diffusive and hydrodynamic magnetotransport around a density perturbation in a two-dimensional electron gas.** P. Shubham Parashar and M. M. Fogler. *Physical Review B* **113**, 155443 (2026). [arXiv:2601.16402](#).
5. **Nanoscale imaging of magnetotransport around a circular p-n junction in graphene.** Z. J. Krebs et al., including P. Shubham Parashar. *Physical Review Letters* **136**, 136301 (2026), Editors' Suggestion. [arXiv:2409.19468](#).
6. **Diagnosing changes in trajectory-geometry tails: endpoint laws, crossovers, and finite-window tests.** P. Shubham Parashar. *Submitted to The European Physical Journal B* (2026).

METHODS AND SELECTED DISTINCTIONS

Methods. Boltzmann/kinetic theory; hydrodynamic closures; magnetotransport and quantum Hall response; asymptotics and worldline methods; finite-resolution inverse/identifiability problems; Python, Mathematica, Kwant, LaTeX.

Distinctions. Bhadra Fellowship (2024); NSF-supported research (2018-2020); UCSD Scholar's Grant (2017); KVPY fellowship (2012-2017).

SELECTED TALKS

Spin-split collapse in higher Landau levels - UC San Diego (2023) | **Vorticity patterns in electronic transport** - Columbia University (2022) | **Magnetotransport across local inhomogeneities** - APS March Meeting (2020)

Updated 11 August 2026